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Chemical compositions and methods of use

Abstract

The present invention is directed to methods for detecting a melanoma, methods for determining whether a melanoma is stable or progressive, methods for evaluating the extent of surgery resection in a subject having a melanoma, and methods for determining a response by a subject having a melanoma to a therapy.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation of U.S. patent application Ser. No. 15/984,935, filed on May 21, 2018, which claims the benefit of and priority to U.S. Provisional Application No. 62/511,058, filed on May 25, 2017. The contents of each of the aforementioned patent application are hereby incorporated by reference in their entireties for all purposes.

INCORPORATION BY REFERENCE OF SEQUENCE LISTING

(1) The instant application contains a Sequence Listing which has been submitted in ASCII format via EFS-Web and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Sep. 28, 2021, is named "LBIO-001_C01US_SeqList.txt" and is about 272,331 bytes in size.

FIELD OF THE INVENTION

(2) The present invention relates to melanoma detection.

BACKGROUND OF THE INVENTION

(3) Melanoma is a common (~24-35/100,000 incidence—US), highly aggressive, skin cancer with an incidence that continues to rise. The most common, cutaneous melanomas, are associated with UV exposure and immune dysregulation. As a group, melanoma is known to carry the highest mutational burden (>10 mutations/Mb). Major mutations include BRAF (~50%), N-Ras (~20%) and NF-1 (~5%), which together, comprise 75% of all mutations. Melanomas are addicted to MAPK pathway activation, regardless of whether tumors exhibit mutations in genes coding for proteins in this pathway. This provides the rationale for targeted therapy e.g., BRAF v600E agents, in this tumor. Other gain-of-function and loss-of-function mutations e.g., in RASopathy genes and amplification of cyclin D1/cdk4 and/or mutation/loss of the tumor suppressor PTEN, also characterize the tumor. This makes melanoma one of the most aggressive and therapy-resistant cancers.

(4) Five-year survival rates range from 95-100% for stage I, 65-93% for stage II, to 41-71% and 9-28% for stage III and IV, respectively. Surgery, immunotherapy and targeted therapies provide the basis for management, with chemotherapy and radiation as adjuncts. Surgery, however, has a critical role in melanoma care (diagnosis, cure and palliation). Sentinel lymph node biopsy has become widespread, as it provides prognostic information. Melanoma, however, lacks a clinically useful non-invasive e.g., blood-based biomarker of disease activity to help guide patient management by providing predictive or prognostic information.

(5) Blood-based factors include lactate dehydrogenase (LDH), detecting mutations in circulating tumor (ct) DNA, measurements of circulating tumor cells (CTCs) and circulating mRNA. LDH is typically used to identify aggressive tumor behavior and predict recurrence but its metrics are very low e.g., 30-50% accurate. It is also non-specific for melanoma. Mutations in target genes, like BRAF, can be detected in the blood in ctDNA but its utility as an indicator of therapeutic efficacy is limited e.g., 45-70% accurate. CTCs do not appear to be an accurate marker in melanomas and there is no consensus as to their clinical utility. Circulating microNAs (miRNA) have been detected but there is no evidence yet for clinical usefulness.

(6) For melanomas, there are no multi-mRNA circulating biomarkers that function as a diagnostic or as a prognostic for disease recurrence. A biomarker that can be used to monitor the efficacy of surgery or drug therapy in melanomas is currently lacking.

SUMMARY OF THE INVENTION

(7) Among other things, disclosed herein is a 28-gene expression tool for melanoma. It has high sensitivity and specificity (>95%) for the detection of melanoma and can differentiate aggressive untreated disease from stable, treated disease.

(8) One aspect of the present disclosure relates to a method for detecting a melanoma in a subject in need thereof, comprising: (1) determining the expression level of at least 29 biomarkers from a test sample from the subject by contacting the test sample with a plurality of agents specific to detect the expression of the at least 29 biomarkers, wherein the at least 29 biomarkers comprise ATL1,

ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, YY2, and at least one housekeeping gene; (2) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of the at least one housekeeping gene, thereby obtaining a normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; (3) inputting each normalized expression level into an algorithm to generate a score; (4) comparing the score with a first predetermined cutoff value; and (5) producing a report, wherein the report identifies the presence of a melanoma in the subject when the score is equal to or greater than the first predetermined cutoff value or identifies the absence of a melanoma in the subject when the score is below the first predetermined cutoff value, wherein the first predetermined cutoff value is 20 on a scale of 0 to 100.

(9) In some embodiments, the method further comprises treating the subject identified as having a melanoma with surgery or drug therapy.

(10) Another aspect of the present disclosure relates to a method for determining whether a melanoma in a subject is stable or progressive, comprising: (1) determining the expression level of at least 29 biomarkers from a test sample from the subject by contacting the test sample with a plurality of agents specific to detect the expression of the at least 29 biomarkers, wherein the at least 29 biomarkers comprise ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, YY2, and at least one housekeeping gene; (2) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of the at least one housekeeping gene, thereby obtaining a normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; (3) inputting each normalized expression level into an algorithm to generate a score; (4) comparing the score with a second predetermined cutoff value; and (5) producing a report, wherein the report identifies that the melanoma is progressive when the normalized expression level is equal to or greater than the second predetermined cutoff value or identifies that the melanoma is stable when the normalized expression level is below the second predetermined cutoff value, wherein the second predetermined cutoff value is 50 on a scale of 0 to 100.

(11) Another aspect of the present disclosure relates to a method for evaluating the extent of surgical resection in a subject having a melanoma, comprising: (1) determining the expression level of at least 29 biomarkers from a test sample from the subject after the surgical resection by contacting the test sample with a plurality of agents specific to detect the expression of the at least 29 biomarkers, wherein the at least 29 biomarkers comprise ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, YY2, and at least one housekeeping gene; (2) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1,

CHP1, DDX55, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of the at least one housekeeping gene, thereby obtaining a normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; (3) inputting each normalized expression level into an algorithm to generate a score; (4) comparing the score with a third predetermined cutoff value; and (5) producing a report, wherein the report identifies that the surgical resection does not remove the entire melanoma when the normalized expression level is equal to or greater than the third predetermined cutoff value or identifies that the surgical resection removes the entire melanoma when the normalized expression level is below the third predetermined cutoff value, wherein the third predetermined cutoff value is 20 on a scale of 0 to 100.

(12) In some embodiments, the report further identifies that the risk of melanoma recurrence is high when the normalized expression level is equal to or greater than the third predetermined cutoff value or identifies that the risk of melanoma recurrence is low when the normalized expression level is below the third predetermined cutoff value.

(13) Yet another aspect of the present disclosure relates to a method for determining a response by a subject having a melanoma to a therapy, comprising: (1) determining a first expression level of at least 28 biomarkers from a first test sample from the subject at a first time point by contacting the first test sample with a plurality of agents specific to detect the expression of the at least 28 biomarkers, wherein the 28 biomarkers comprise ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; (2) determining a second expression level of the at least 28 biomarkers from a second test sample from the subject at a second time point by contacting the second test sample with a plurality of agents specific to detect the expression of the at least 28 biomarkers, wherein the second time point is after the first time point and after the administration of the therapy to the subject; (3) comparing the first expression level with the second expression level; and (4) producing a report, wherein the report identifies that the subject is responsive to the therapy when the second expression level is significantly decreased as compared to the first expression level.

(14) In some embodiments, the first time point is prior to the administration of the therapy to the subject. In some embodiments, the first time point is after the administration of the therapy to the subject. In some embodiments, the therapy comprises an immunotherapy or a targeted therapy (e.g., a BRAF inhibitor).

(15) In some embodiments, the at least one housekeeping gene is selected from the group consisting of ALG9, SEPN, YWHAQ, VPS37A, PRRC2B, DOPEY2, NDUFB11, ND4, MRPL19, PSMC4, SF3A1, PUM1, ACTB, GAPD, GUSB, RPLP0, TFRC, MORF4L1, 18S, PPIA, PGK1, RPL13A, B2M, YWHAZ, SDHA, HPRT1, TOX4, and TPT1.

(16) In some embodiments, the at least one housekeeping gene comprises TOX4 and TPT1.

(17) In some embodiments, the normalized expression level is obtained by: (1) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of TOX4, thereby obtaining a first normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5,

SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; (2) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of TPT1, thereby obtaining a second normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; and (3) averaging the first normalized expression level and the second normalized expression level to obtain the normalized expression level.

(18) In some embodiments, the method can have a specificity, sensitivity, and/or accuracy of at least 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%. In some embodiments, the method has a sensitivity of greater than 90%. In some embodiments, the method has a specificity of greater than 90%.

(19) In some embodiments, the biomarker is RNA, cDNA, or protein. When the biomarker is RNA, the RNA can be reverse transcribed to produce cDNA, and the produced cDNA expression level is detected. In some embodiments, the expression level of the biomarker is detected by forming a complex between the biomarker and a labeled probe or primer. When the biomarker is RNA or cDNA, the RNA or cDNA can be detected by forming a complex between the RNA or cDNA and a labeled nucleic acid probe or primer. When the biomarker is protein, the protein can be detected by forming a complex between the protein and a labeled antibody. In some embodiments, the label is a fluorescent label.

(20) In some embodiments, the test sample is blood, serum, plasma, or neoplastic tissue.

(21) In some embodiments, the first predetermined cutoff value can be derived from a plurality of reference samples obtained from subjects free of a neoplastic disease. The second predetermined cutoff value can be derived from a plurality of reference samples obtained from subjects whose melanomas are being adequately controlled by therapies like immune therapy. The third predetermined cutoff value can be derived from a plurality of reference samples obtained from subjects whose melanomas have been completely removed by surgery and they are considered “disease free.” In some embodiments, each reference sample can be blood, serum, plasma, or non-neoplastic tissue.

(22) In some embodiments, the subject in need thereof is a subject diagnosed with a melanoma, a subject having at least one melanoma symptom, or a subject having a predisposition or familial history for developing a melanoma. In some embodiments, the subject is a human.

(23) In some embodiments, the algorithm is XGB, RF, glmnet, cforest, CART, treebag, knn, nnet, SVM-radial, SVM-linear, NB, NNET, or mlp.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a graph showing visualization of the melanoma score as a system of three contributors to the clinical picture—Control, Response, and Progression. Samples towards each of the corner represent pure representations of each clinical group. Samples in the middle are in the area of both algorithmic and clinical uncertainty.

(2) FIG. 2 is a graph showing the metrics for the test in the test set ranged from 87-100%.

(3) FIGS. 3A-3C are a set of graphs showing the evaluation of the circulating melanoma gene test (Melanomx) in test set 2. Values were significantly higher in melanoma samples than in controls (FIG. 3A). Patients who were responding to therapy had values similar to controls. Receiver operator curve analysis of test set 2 identifying the AUC for differentiating melanoma from

- control was >0.95 (FIG. 3B). The metrics for the test ranged from 78-92% (FIG. 3C).
- (4) FIGS. 4A-4B are a set of graphs showing the effect of surgery on the Melanomx. Levels were significantly decreased by surgery ($p<0.0001$) (FIG. 4A). Values in the NED (no evidence of disease after surgery) group were significantly lower than in those with residual disease after surgery ($p=0.0007$) (FIG. 4B).
- (5) FIG. 5 is a graph showing the effect of therapy on the Melanomx score. Levels were significantly decreased by immunotherapy (ipilimumab) or a BRAF inhibitor (Vemurafenib).
- (6) FIGS. 6A-6B are a set of graphs showing Melanomx score in 3 different melanoma cell lines. FIG. 6A identifies the cell lines demonstrate elevated expression—Melanomx score ranging from 40 (A375) to 95 (Hs294). FIG. 6B identifies that spiking these cells into blood from a subject that does not have a melanoma, resulted in detectable gene expression and scores. A minimum of 1 cell/ml of blood could be consistently identified.
- (7) FIGS. 7A-7B are a set of graphs showing expression in tumor tissue and its correlation with blood samples collected at the same time. In FIG. 7A, the Melanomx score ranged 40-97 in melanoma tumor tissue. In contrast, normal epithelium exhibited values <20 . In FIG. 7B, gene expression in tumor tissue is compared to matched blood samples. This is highly concordant (correlation ~ 0.80).

DETAILED DESCRIPTION OF THE INVENTION

- (8) The details of the invention are set forth in the accompanying description below. Although methods and materials similar or equivalent to those described herein can be used in the practice or testing of the present invention, illustrative methods and materials are now described. Other features, objects, and advantages of the invention will be apparent from the description and from the claims. In the specification and the appended claims, the singular forms also include the plural unless the context clearly dictates otherwise. Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. All patents and publications cited in this specification are incorporated herein by reference in their entireties.
- (9) Early signs of melanoma include changes to the shape or color of existing moles or, in the case of nodular melanoma, the appearance of a new lump anywhere on the skin. At later stages, the mole may itch, ulcerate or bleed. Visual inspection is the most common diagnostic technique. Melanoma can be divided into the following types: lentigo maligna, lentigo maligna melanoma, superficial spreading melanoma, acral lentiginous melanoma, mucosal melanoma, nodular melanoma, polypoid melanoma, and desmoplastic melanoma.
- (10) Measurements of circulating melanoma transcripts—the Melanomx—identify melanomas and decreases in the Melanomx score correlate with the efficacy of therapeutic interventions such as immunotherapy and targeted therapy. Targeted gene expression profile of RNA can be isolated from the peripheral blood of patients with melanoma. This expression profile is evaluated in an algorithm and converted to an output (prediction). It can identify active disease, provide an assessment of treatment responses, or predict risk of relapse, in conjunction with standard clinical assessment and imaging.
- (11) In one aspect, the present disclosure provides a method for detecting a melanoma in a subject in need thereof, including: (1) determining the expression level of at least 29 biomarkers from a test sample from the subject by contacting the test sample with a plurality of agents specific to detect the expression of the at least 29 biomarkers, wherein the at least 29 biomarkers comprise ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, YY2, and at least one housekeeping gene; (2) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127,

LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of the at least one housekeeping gene, thereby obtaining a normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; (3) inputting each normalized expression level into an algorithm to generate a score; (4) comparing the score with a first predetermined cutoff value; and (5) producing a report, wherein the report identifies the presence of a melanoma in the subject when the score is equal to or greater than the first predetermined cutoff value or identifies the absence of a melanoma in the subject when the score is below the first predetermined cutoff value, wherein the first predetermined cutoff value is 20 on a scale of 0 to 100.

(12) In some embodiments, the at least one housekeeping gene is selected from the group consisting of ALG9, SEPN, YWHAQ, VPS37A, PRRC2B, DOPEY2, NDUFB11, ND4, MRPL19, PSMC4, SF3A1, PUM1, ACTB, GAPD, GUSB, RPLP0, TFRC, MORF4L1, 18S, PPIA, PGK1, RPL13A, B2M, YWHAZ, SDHA, HPRT1, TOX4, and TPT1.

(13) In some embodiments, the at least one housekeeping gene comprises TOX4 and TPT1. In some embodiments, the normalized expression level is obtained by: (1) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of TOX4, thereby obtaining a first normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; (2) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of TPT1, thereby obtaining a second normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; and (3) averaging the first normalized expression level and the second normalized expression level to obtain the normalized expression level.

(14) Among the provided methods are those that are able to classify or detect a melanoma. In some embodiments, the provided methods can identify or classify a melanoma in a human blood sample. In some examples, the methods can provide such information with a specificity, sensitivity, and/or accuracy of at least 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%.

(15) The agents can be any agents for detection of the biomarkers, and typically are isolated polynucleotides or isolated polypeptides or proteins, such as antibodies, for example, those that specifically hybridize to or bind to the at least 29 biomarkers.

(16) The biomarker can be RNA, cDNA, or protein. When the biomarker is RNA, the RNA can be reverse transcribed to produce cDNA (such as by RT-PCR), and the produced cDNA expression level is detected. The expression level of the biomarker can be detected by forming a complex between the biomarker and a labeled probe or primer. When the biomarker is RNA or cDNA, the RNA or cDNA detected by forming a complex between the RNA or cDNA and a labeled nucleic acid probe or primer. The complex between the RNA or cDNA and the labeled nucleic acid probe or primer can be a hybridization complex.

(17) When the biomarker is protein, the protein can be detected by forming a complex between the protein and a labeled antibody. The label can be any label, for example a fluorescent label,

chemiluminescence label, radioactive label, etc. The protein level can be measured by methods including, but not limited to, immunoprecipitation, ELISA, Western blot analysis, or immunohistochemistry using an agent, e.g., an antibody, that specifically detects the protein encoded by the gene.

(18) In some embodiments, the methods are performed by contacting the test sample with one of the provided agents, more typically with a plurality of the provided agents, for example, a set of polynucleotides that specifically bind to the at least 29 biomarkers. In some embodiments, the set of polynucleotides includes DNA, RNA, cDNA, PNA, genomic DNA, or synthetic oligonucleotides. In some embodiments, the methods include the step of isolating RNA from the test sample prior to detection, such as by RT-PCR, e.g., QPCR. Thus, in some embodiments, detection of the melanoma biomarkers, such as expression levels thereof, includes detecting the presence, absence, or amount of RNA. In one example, the RNA is detected by PCR or by hybridization.

(19) In some embodiments, the polynucleotides include sense and antisense primers, such as a pair of primers that is specific to each of the at least 29 biomarkers. In one aspect of this embodiment, the detection of the at least 29 biomarkers is carried out by PCR, typically quantitative or real-time PCR. For example, in one aspect, detection is carried out by producing cDNA from the test sample by reverse transcription; then amplifying the cDNA using the pairs of sense and antisense primers that specifically hybridize to the panel of at least 28 biomarkers, and detecting products of the amplification.

(20) The test sample can be any biological fluid obtained from the subject. Preferably, the test sample is blood, serum, plasma or neoplastic tissue.

(21) The first predetermined cutoff value can be derived from a plurality of reference samples obtained from subjects free of a neoplastic disease. Each reference sample can be any biological fluid obtained from a subject not having, showing symptoms of or diagnosed with a neoplastic disease. In some embodiments, the reference sample is blood, serum, plasma, or non-neoplastic tissue.

(22) The subject in need thereof can be a subject diagnosed with a melanoma, a subject having at least one melanoma symptom or a subject having a predisposition or familial history for developing a melanoma. The subject can be any mammal. Preferably, the subject is human. The terms “subject” and “patient” are used interchangeably herein.

(23) The score is the Melanomx score, which has a scale of 0 to 100. The Melanomx score is the product of a classifier built from predictive classification algorithms, e.g., XGB, RF, glmnet, cforest, CART, treebag, knn, nnet, SVM-radial, SVM-linear, NB, NNET, or mlp. The algorithm analyzes the data (i.e., expression levels) and then assigns a score.

(24) The method can further include treating the subject identified as having a melanoma with surgery, drug therapy, radiation therapy, or a combination thereof. The drug therapy can be an immunotherapy, a targeted therapy, a chemotherapy, or a combination thereof. In some embodiments, the drug therapy includes an immunotherapy. Examples of immunotherapies for treating a melanoma include, but are not limited to, Imlygic (T-VEC), Yervoy in combination with Opdivo, Opdivo (nivolumab), Keytruda (pembrolizumab), Yervoy (ipilimumab), Interleukin-2 (IL-2), and Interferon alpha 2-b. In some embodiments, the drug therapy includes a targeted therapy such as a BRAF inhibitor. Examples of targeted therapies for treating a melanoma include, but are not limited to, Zelboraf in combination with Cotellic (cobimetinib), Tafenlar in combination with Mekinist, Tafenlar (dabrafenib), Mekinist (trametinib), and Zelboraf (vemurafenib). In some embodiments, the drug therapy includes a chemotherapy. In some embodiments, the chemotherapy includes dacarbazine.

(25) Accordingly, the present disclosure provides a method of treating melanoma in a human subject in need thereof, the method comprising: a) determining the expression level of 29 biomarkers from a biological fluid sample from the subject by contacting the test sample with a

plurality of agents specific to detect the expression of the 29 biomarkers, wherein the 29 biomarkers comprise each of atlastin GTPase 1 (ATL1), ATPase H⁺ transporting V0 subunit d1 (ATP6V0D), chromosome 1 open reading frame 21 (C1ORF21), CASP8 and FADD like apoptosis regulator (CFLAR), CFLAR antisense RNA 1 (CFLAR-AS1), calcineurin like EF-hand protein 1 (CHP1), DEAD-box helicase 55 (DDX55), dystrophin (DMD), DnaJ heat shock protein family (Hsp40) member C9 (DNAJC9), enolase superfamily member 1 (ENOSF1), Fanconi anemia complementation group L (FANCL), Holliday junction recognition protein (HJURP), major histocompatibility complex, class II, DO alpha (HLA-DOA), major histocompatibility complex, class II, DR alpha (HLA-DRA), heterogeneous nuclear ribonucleoprotein A3 pseudogene 1 (HNRNPA3P1), interleukin 23 subunit alpha (IL23A), IQ motif containing GTPase activating protein 1 (IQGAP1), NFYC pseudogene (LOC494127), uncharacterized LOC646471 (LOC646471), loss of heterozygosity, 12, chromosomal region 2 (non-protein coding) (LOH12CR), PBX homeobox interacting protein 1 (PBXIP1), ring finger protein 5 (RNF5), SERTA domain containing 2 (SERTAD2), solute carrier family 35 member G5 (SLC35G5), spermatogenesis associated serine rich 2 like (SPATS2L), tudor domain containing 7 (TDRD7), TXK tyrosine kinase (TXK), YY2 transcription factor (YY2), and at least one housekeeping gene; b) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of the at least one housekeeping gene, thereby obtaining a normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; c) inputting each normalized expression level into an algorithm to generate a score, wherein the algorithm is the product of a trained classifier built from at least one predictive classification algorithm; d) determining that the human subject has melanoma by determining that the score is greater than or equal to a first predetermined cutoff value, wherein the first predetermined cutoff value is 20 on a scale of 0 to 100; and e) administering to the human subject at least one therapeutically effective amount of at least one of:

(26) ##STR00001##

or any combination thereof.

(27) The present disclosure also provides a method for determining whether a melanoma in a subject is stable or progressive, including: (1) determining the expression level of at least 29 biomarkers from a test sample from the subject by contacting the test sample with a plurality of agents specific to detect the expression of the at least 29 biomarkers, wherein the at least 29 biomarkers comprise ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, YY2, and at least one housekeeping gene; (2) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of the at least one housekeeping gene, thereby obtaining a normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; (3) inputting each normalized expression level into an algorithm to generate a score; (4) comparing the score with a second predetermined cutoff value; and (5) producing a report, wherein the report identifies that the melanoma is progressive when the

normalized expression level is equal to or greater than the second predetermined cutoff value or identifies that the melanoma is stable when the normalized expression level is below the second predetermined cutoff value, wherein the second predetermined cutoff value is 50 on a scale of 0 to 100.

(28) The second predetermined cutoff value can be derived from a plurality of reference samples obtained from subjects whose melanomas are being adequately controlled by therapies like immune therapy.

(29) Surgical resection is a procedure that removes melanoma tissues from the subject in need thereof. The present disclosure also provides a method for evaluating the extent of surgical resection in a subject having a melanoma, including: (1) determining the expression level of at least 29 biomarkers from a test sample from the subject after the surgical resection by contacting the test sample with a plurality of agents specific to detect the expression of the at least 29 biomarkers, wherein the at least 29 biomarkers comprise ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, YY2, and at least one housekeeping gene; (2) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of the at least one housekeeping gene, thereby obtaining a normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; (3) inputting each normalized expression level into an algorithm to generate a score; (4) comparing the score with a third predetermined cutoff value; and (5) producing a report, wherein the report identifies that the surgical resection does not remove the entire melanoma when the normalized expression level is equal to or greater than the third predetermined cutoff value or identifies that the surgical resection removes the entire melanoma when the normalized expression level is below the third predetermined cutoff value, wherein the third predetermined cutoff value is 20 on a scale of 0 to 100.

(30) The third predetermined cutoff value can be derived from a plurality of reference samples obtained from subjects whose melanoma disease has been completely removed by surgery and they are considered “disease free.”

(31) When it is determined that the surgical resection does not remove the entire melanoma, the subject is at risk of melanoma recurrence. Accordingly, in some embodiments, the report further identifies that the risk of melanoma recurrence is high when the normalized expression level is equal to or greater than the third predetermined cutoff value or identifies that the risk of melanoma recurrence is low when the normalized expression level is below the third predetermined cutoff value.

(32) The present disclosure also provides a method for determining a response by a subject having a melanoma to a therapy, comprising: (1) determining a first expression level of at least 28 biomarkers from a first test sample from the subject at a first time point by contacting the first test sample with a plurality of agents specific to detect the expression of the at least 28 biomarkers, wherein the 28 biomarkers comprise ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; (2) determining a second expression level of the at least 28 biomarkers from a second test sample from the subject at a second time point by contacting the second test sample with a plurality of agents specific to detect the expression of the at least 28

biomarkers, wherein the second time point is after the first time point and after the administration of the therapy to the subject; (3) comparing the first expression level with the second expression level; and (4) determining that the subject is responsive to the therapy when the second expression level is significantly decreased as compared to the first expression level.

(33) In some embodiments, the methods can predict treatment responsiveness to, or determine whether a patient has become clinically stable following, or is responsive or non-responsive to, a melanoma treatment, such as a surgical intervention or drug therapy (for example, an immunotherapy or targeted therapy). In some cases, the methods can do so with a specificity, sensitivity, and/or accuracy of at least 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%. In some cases, it can differentiate between treated and untreated melanoma with a specificity, sensitivity, and/or accuracy of at least 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%.

(34) In some embodiments, the first and second test samples can be of the same type. In some embodiments, the first and second test samples can be of different types.

(35) In some embodiments, the therapy can be a drug therapy. The drug therapy can be an immunotherapy, a targeted therapy, a chemotherapy, or a combination thereof. In some embodiments, the therapy can be a radiation therapy.

(36) In some embodiments, the first time point is prior to the administration of the therapy to the subject. In some embodiments, the first time point is after the administration of the therapy to the subject. The second time point can be a few days, a few weeks, or a few months after the first time point. For example, the second time point can be at least 1 day, at least 7 days, at least 14 days, at least 30 days, at least 60 days, or at least 90 days after the first time point.

(37) In some embodiments, the second expression level is significantly decreased as compared to the first expression level when the second expression level is at least 10% less than the first expression level. In some embodiments, the second expression level is significantly decreased as compared to the first expression level when the second expression level is at least 20% less than the first expression level. In some embodiments, the second expression level is significantly decreased as compared to the first expression level when the second expression level is at least 30% less than the first expression level. In some embodiments, the second expression level is significantly decreased as compared to the first expression level when the second expression level is at least 40% less than the first expression level. In some embodiments, the second expression level is significantly decreased as compared to the first expression level when the second expression level is at least 50% less than the first expression level. In some embodiments, the second expression level is significantly decreased as compared to the first expression level when the second expression level is at least 60% less than the first expression level. In some embodiments, the second expression level is significantly decreased as compared to the first expression level when the second expression level is at least 70% less than the first expression level. In some embodiments, the second expression level is significantly decreased as compared to the first expression level when the second expression level is at least 80% less than the first expression level. In some embodiments, the second expression level is significantly decreased as compared to the first expression level when the second expression level is at least 90% less than the first expression level.

(38) In some embodiments, the method further comprises determining a third expression level of the at least 28 biomarkers from a third test sample from the subject at a third time point by contacting the third test sample with a plurality of agents specific to detect the expression of the at least 28 biomarkers, wherein the third time point is after the second time point. The method can further comprise creating a plot showing the trend of the expression level change.

(39) The present disclosure also provides an assay comprising: (1) determining the expression level of biomarkers consisting essentially of the following 30 biomarkers from a test sample from a patient diagnosed of a melanoma or a subject suspected of having a melanoma: ATL1, ATP6V0D,

C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, YY2, TOX4, and TPT1, wherein the expression level is measured by contacting the test sample with a plurality of agents specific to detect the expression of the 30 biomarkers; (2) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of TOX4, thereby obtaining a first normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; (3) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of TPT1, thereby obtaining a second normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; (4) averaging the first normalized expression level and the second normalized expression level to obtain a normalized expression level for each biomarker; (5) inputting each normalized expression level into an algorithm to generate a score; and (6) comparing the score with a first predetermined cutoff value.

(40) The present disclosure also provides an assay comprising: (1) determining the expression level of biomarkers consisting of the following 30 biomarkers from a test sample from a patient diagnosed of a melanoma or a subject suspected of having a melanoma: ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, YY2, TOX4, and TPT1, wherein the expression level is measured by contacting the test sample with a plurality of agents specific to detect the expression of the 30 biomarkers; (2) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of TOX4, thereby obtaining a first normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; (3) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of TPT1, thereby obtaining a second normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; (4) averaging the first normalized expression level and the second normalized expression level to obtain a normalized expression level for each biomarker; (5) inputting each normalized expression level into an algorithm to generate a score; and (6) comparing the score with a first predetermined cutoff value.

(41) The sequence information of the melanoma biomarkers and housekeepers is shown in Table 1.

(42)	TABLE-000001	TABLE	1	Melanoma	Housekeeper/	Sequence
Information	SEQ	Gene	RefSeq	ID	Name	Accession
						Sequence NO: ATL1 NM_001127713.1
AGTTTGAGGC	TGCACTGCAC	ACCTGCACAA	CCTGCCTTCT	1	ACTTAGTTCT	
TCTGAGACAT	TTCTGAAAGT	CTGAATTCCT	AGGACTGCTC	AATGACCTTT		
GTCCCTGTTG	GGCACACGCA	GTGTCTCATC	GCTGGTATTG	CACCTTTAAT		
GAGACCAGGA	GTTCCGCAAA	AGTAAAACAA	AGGGGTCACA	GA CTGGTTCT		
GGTTCTACCA	CTTCCTCAGT	ATGTGCTCTG	GGACAAAACA	GCATATTTGG		
CAACATCTCG	GTGTCCTTAT	CTGCAGCTTG	AAGAGGCGCC	ACAGCAACAT		
CCTCAGAGTC	TGAGCGAACT	GCGCCCAGCG	CGGGCACGGA	GCCTCCCACC		
GCCAGCAACC	TGCGGCCCCG	GAGAAGGCAG	CGAGCGCAGT	GACAGCGCCT		
CACCGCCACC	AGCTCCTGGA	CCACCATGGC	CAAGAACCGC	AGGGACAGAA		
ACAGTTGGGG	TGGATTTTCG	GAAAAGACAT	ATGAATGGAG	CTCAGAAGAG		
GAGGAGCCAG	TGAAAAAGGC	AGGACCAGTC	CAAGTCCTCA	TTGTCAAAGA		
TGACCATTCC	TTTGAGTTAG	ATGAAACTGC	ATTAAATCGG	ATCCTTCTCT		
CGGAGGCTGT	CAGAGACAAG	GAGGTTGTTG	CTGTATCTGT	TGCTGGAGCA		
TTTAGAAAAG	GAAAATCATT	CCTGATGGAC	TTCATGTTGA	GATACATGTA		
CAACCAGGAA	TCAGTTGATT	GGGTTGGAGA	CTACAATGAA	CCATTGACTG		
GTTTTTCATG	GAGAGGTGGA	TCTGAGCGAG	AGACCACAGG	AATTCAGATA		
TGGAGTGAAA	TCTTCCTTAT	CAATAAACCT	GATGGTAAAA	AGGTTGCAGT		
GTTATTGATG	GATACTCAGG	GAACCTTTGA	TAGTCAGTCA	ACTTTGAGAG		
ATTCAGCCAC	AGTATTTGCC	CTTAGCACAA	TGATCAGCTC	AATACAGGTA		
TATAACTTAT	CCCAAATGT	CCAGGAGGAT	GATCTTCAGC	ACCTCCAGCT		
TTTCACTGAG	TATGGCAGAC	TGGCAATGGA	GGAAACATTC	CTGAAGCCAT		
TTCAGAGTCT	GATATTTCTT	GTTGAGACT	GGAGTTTCCC	ATACGAATTT		
TCATATGGAG	CCGATGGTGG	TGCCAAATTC	TTGGAAAAAC	GCCTCAAGGT		
CTCAGGGAAC	CAGCATGAAG	AACTACAGAA	CGTCAGAAAA	CACATCCATT		
CCTGTTTCAC	CAACATTTCC	TGTTTTCTGC	TACCTCATCC	TGGCTTAAAA		
GTAGCTACCA	ATCCAAACTT	TGATGGAAAA	TTGAAAGAAA	TAGATGATGA		
ATTCATCAAA	AACTTGAAAA	TACTGATTCC	TTGGCTACTT	AGTCCCGAGA		
GCCTAGATAT	TAAAGAGATC	AATGGGAATA	AAATCACCTG	CCGGGGTCTG		
GTGGAGTACT	TCAAGGCTTA	TATAAAGATC	TATCAAGGTG	AAGAATTACC		
ACATCCCAA	TCCATGTTAC	AGGCCACAGC	AGAAGCTAAC	AATTTAGCAG		
CCGTGGCAAC	TGCCAAGGAC	ACATACAACA	AAAAAATGGA	AGAGATTTGT		
GGTGGTGACA	AACCATTTCT	GGCCCCAAT	GA CTTGCAGA	CCAAACACCT		
GCAACTTAAG	GAAGAATCTG	TGAAGCTATT	CCGAGGGGTG	AAGAAGATGG		
GTGGGGAAGA	ATTTAGCCGG	CGTTACCTGC	AGCAGTTGGA	GAGTGAAATA		
GATGAACTTT	ACATCCAATA	TATCAAGCAC	AATGATAGCA	AAAATATCTT		
CCATGCAGCT	CGTACCCCAG	CCACACTGTT	TGTAGTCATC	TTTATCACAT		
ATGTGATTGC	TGGTGTGACT	GGATTCATTG	GTTTGGACAT	CATAGCTAGC		
CTATGCAATA	TGATAATGGG	ACTGACCCTT	ATCACCTGT	GCACTTGGGC		
ATATATCCGG	TACTCTGGAG	AATACCGAGA	GCTGGGAGCT	GTAATAGACC		
AGGTGGCTGC	AGCTCTGTGG	GACCAGGCTT	TGTACAAGCT	TTACAGTGCA		
GCAGCAACCC	ACAGACATCT	GTATCATCAA	GCTTTCCCTA	CACCAAAGTC		
GGAATCTACT	GAACAATCAG	AAAAGAAAAA	AATGTAATGC	AAATTTTAAG		
AAATACAGGT	GCATGACCAA	TTGTCAATTA	AATATTCAGT	TTTATGTCTC		
CATGCAAACA	TTCAAAGTGC	TTCCATCAGA	ACGGAGTAAA	ATACTAAACA		
CCTCTGAAGA	CTGCAAACCTG	GATTAGTTCT	TTTACTTCAG	TGTTTAATAA		
GCAGATGTAT	GTATGCATGG	TTATACTATT	TTGTTAACAT	GTACAATTTC		
CTGATTTTTT	TTCAAAAATG	CTGTTATAAA	GTATTTGTCT	ATTTATGATA		
ACAGTACACG	TGTTCTGCTT	GAATTTACTA	AATTCTACTA	CTGGGTTATA		

ATTAATCAT GTGATATCC ACCTTGTGATG ATGCTCATTTAATTCTACA
GAAAAAATTT TAAATTATTT CACATTAGCC ATTTGTTAAA ACACAGCATC
ATAACTCAGC AGGCTGGATT TAATCTGTAT CATCTTATAT ATATCACAAT
CTTATTTTTTAAGCACATTTT AGAGTTCCTT AGTTGCTTTA TCAAAAACCA
GATATTGCTT TTACATGGTT TAATAGAATA TAAACCTCTT GATAAAAAAT
GCACAAAAAA TCACTTTGTA TATGTGAGTT TCACTGCATT GTATATTTTT
TCATTTGGTA CACAAAGAAT GTATTCTTCA TAGGTTTATT CTTTTAATAT
GTGAACTATT ATTAAAGTTT ACTCTGGTTC CTAAGATTAA AAACAAATGC
TIACTGAATT TGAAAAAAA AATP6V0D NM_004691.4 ACTTGACAGC
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CCGGTCCTGG TTCTCCGGTC CCGCCGCTCC CGCAGCAGCC ATGTCGTTCT
TCCCGGAGCT TIACTTTAAC GTGGACAATG GCTACTTGGA GGGACTGGTG
CGCGGCCTGA AGGCCGGGGT GCTCAGCCAG GCCGACTACC TCAACCTGGT
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ATGGTAACTT CCTGGCCAAC GAGGCATCAC CTCTGACGGT GTCAGTCATC
GATGACCGGC TCAAGGAGAA GATGGTGGTG GAGTTCCGCC ACATGAGGAA
CCATGCCTAT GAGCCACTCG CCAGCTTCCT AGACTTCATT ACTTACAGTT
ACATGATCGA CAACGTGATC CTGCTCATCA CAGGCACGCT GCACCAGCGC
TCCATCGCTG AGCTCGTGCC CAAGTGCCAC CCACTAGGCA GCTTCGAGCA
GATGGAGGCC GTGAACATTG CTCAGACACC TGCTGAGCTC TACAATGCCA
TTCTGGTGGA CACGCCTCTT GCGGCTTTTT TCCAGGACTG CATTTCAGAG
CAGGACCTTG ACGAGATGAA CATCGAGATC ATCCGCAACA CCCTCTACAA
GGCCTACCTG GAGTCCTTCT ACAAGTTCTG CACCCTACTG GCGGGGACTA
CGGCTGATGC CATGTGCCCC ATCCTGGAGT TTGAAGCAGA CCGCCGCGCC
TTCATCATCA CCATCAATTC TTTCGGCACA GAGCTGTCCA AAGAGGACCG
TGCCAAGCTC TTTCCACACT GTGGGCGGCT CTACCCTGAG GGCCTGGCGC
AGCTGGCTCG GGCTGACGAC TATGAACAGG TCAAGAACGT GGCCGATTAC
TACCCGGAGT ACAAGCTGCT CTTCGAGGGT GCAGGTAGCA ACCCTGGAGA
CAAGACGCTG GAGGACCGAT TCTTTGAGCA CGAGGTAAAG CTGAACAAGT
TGGCCTTCCT GAACCAGTTC CACTTTGGTG TCTTCTATGC CTTCGTGAAG
CTCAAGGAGC AGGAGTGTCG CAACATCGTG TGGATCGCTG AATGTATCGC
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ACAGTTTTGA CATGTTATAC TTGCGCATAG ATCCAAGCGT TTCTTGGGAA
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ATAGATCCAA GAAATGGGGT GGTTGAGTGG GTCCGCACGA AATGCTTGAT
TATGTCAGCA ACACCCAACA CTGTCTGTTT TCCATTTGTT GGTTTTAATC
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GAAAGGATCT AAAACAAAAA TATTTTGCCT TTTTTTCCCC ACGTGTATCT
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CGTGCTGGTG CACTCTGCTA GTTGTTATCT CTGTAGGGCT CAGGAAGCTG
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TTTTAAGGTT AAAAGTCTTG ACCTTCATGG GGGTCTGGGA CAATCCGATC
TCCAAGCATG GAGGAAAGGC AATGCCTGGA CCACTGACTT GCATTGAAAT
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GATCTGCATC TTCCAGGTCC ATTTAAGCAC TGAAACTAGA TGCAAATCTC
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TGAAGGCTTC ATTCTGATAG TGTCTGCCCT CTCTACCCTG ATTCGCCCC
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ACTCTAAAGG GAGAGAATAG CCCCTGTGTC CTGGCATTTC AGTCTAGACC
TTCAAGGACT GTTCTCTCTT GACAGGCAAG CAAGCAAAGA AAGTTTTGCA
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GATTCGGATC TATTTTCTTA ACACACATAC ACATACACGC GCATACATAC
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TAGGAAAGTA ATAGTATAAC TATAGGGATA CCGAAACAGG AAAAACCAGC
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CGTCCCGCCA GCCATTCGCT CCTACTGATG AGACAAGATG TGGTGATGAC
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TTGACCTAAA AATCAAATGT AATCTTTGTA TTGTGTTACA TCAAATCCA
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HPRT1 NM_000194.2 GGCGGGGCCT GCTTCTCCTC AGCTTCAGGC GGCTGCGACG
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AGCCAGACTT TGTTGGATTT GAAATTCCAG ACAAGTTTGT TGTAGGATAT

GCCCTTGA CT ATAATGAATA CTCAGGGAT TTGAATCATG TTTGTGTCAT
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 AATCAATAAA TTCCCAAAAA TTAAAAAAA AAAAAAAAAA AAAAA

Definitions

(43) The articles “a” and “an” are used in this disclosure to refer to one or more than one (i.e., to at least one) of the grammatical object of the article. By way of example, “an element” means one element or more than one element.

(44) The term “and/or” is used in this disclosure to mean either “and” or “or” unless indicated otherwise.

(45) As used herein, the terms “polynucleotide” and “nucleic acid molecule” are used interchangeably to mean a polymeric form of nucleotides of at least 10 bases or base pairs in length, either ribonucleotides or deoxynucleotides or a modified form of either type of nucleotide, and is meant to include single and double stranded forms of DNA. As used herein, a nucleic acid molecule or nucleic acid sequence that serves as a probe in a microarray analysis preferably comprises a chain of nucleotides, more preferably DNA and/or RNA. In other embodiments, a nucleic acid molecule or nucleic acid sequence comprises other kinds of nucleic acid structures such as for instance a DNA/RNA helix, peptide nucleic acid (PNA), locked nucleic acid (LNA) and/or a ribozyme. Hence, as used herein the term “nucleic acid molecule” also encompasses a chain comprising non-natural nucleotides, modified nucleotides and/or non-nucleotide building blocks which exhibit the same function as natural nucleotides.

(46) As used herein, the terms “hybridize,” “hybridizing,” “hybridizes,” and the like, used in the context of polynucleotides, are meant to refer to conventional hybridization conditions, such as hybridization in 50% formamide/6×SSC/0.1% SDS/100 µg/ml ssDNA, in which temperatures for hybridization are above 37 degrees and temperatures for washing in 0.1×SSC/0.1% SDS are above 55 degrees C., and preferably to stringent hybridization conditions.

(47) As used herein, the term “normalization” or “normalizer” refers to the expression of a differential value in terms of a standard value to adjust for effects which arise from technical variation due to sample handling, sample preparation, and measurement methods rather than biological variation of biomarker concentration in a sample. For example, when measuring the expression of a differentially expressed protein, the absolute value for the expression of the protein can be expressed in terms of an absolute value for the expression of a standard protein that is substantially constant in expression.

(48) The terms “diagnosis” and “diagnostics” also encompass the terms “prognosis” and “prognostics”, respectively, as well as the applications of such procedures over two or more time points to monitor the diagnosis and/or prognosis over time, and statistical modeling based thereupon. Furthermore, the term diagnosis includes: a. prediction (determining if a patient will likely develop aggressive disease (hyperproliferative/invasive)), b. prognosis (predicting whether a patient will likely have a better or worse outcome at a pre-selected time in the future), c. therapy selection, d. therapeutic drug monitoring, and e. relapse monitoring.

(49) The term “providing” as used herein with regard to a biological sample refers to directly or indirectly obtaining the biological sample from a subject. For example, “providing” may refer to the act of directly obtaining the biological sample from a subject (e.g., by a blood draw, tissue biopsy, lavage and the like). Likewise, “providing” may refer to the act of indirectly obtaining the biological sample. For example, providing may refer to the act of a laboratory receiving the sample from the party that directly obtained the sample, or to the act of obtaining the sample from an archive.

(50) “Accuracy” refers to the degree of conformity of a measured or calculated quantity (a test reported value) to its actual (or true) value. Clinical accuracy relates to the proportion of true outcomes (true positives (TP) or true negatives (TN) versus misclassified outcomes (false positives (FP) or false negatives (FN)), and may be stated as a sensitivity, specificity, positive predictive values (PPV) or negative predictive values (NPV), or as a likelihood, odds ratio, among other measures.

(51) The term “biological sample” as used herein refers to any sample of biological origin potentially containing one or more biomarkers. Examples of biological samples include tissue, organs, or bodily fluids such as whole blood, plasma, serum, tissue, lavage or any other specimen used for detection of disease.

(52) The term “subject” as used herein refers to a mammal, preferably a human.

(53) “Treating” or “treatment” as used herein with regard to a condition may refer to preventing the condition, slowing the onset or rate of development of the condition, reducing the risk of developing the condition, preventing or delaying the development of symptoms associated with the condition, reducing or ending symptoms associated with the condition, generating a complete or partial regression of the condition, or some combination thereof.

(54) Biomarker levels may change due to treatment of the disease. The changes in biomarker levels may be measured by the present disclosure. Changes in biomarker levels may be used to monitor the progression of disease or therapy.

(55) “Altered”, “changed” or “significantly different” refer to a detectable change or difference from a reasonably comparable state, profile, measurement, or the like. Such changes may be all or none. They may be incremental and need not be linear. They may be by orders of magnitude. A change may be an increase or decrease by 1%, 5%, 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, 99%, 100%, or more, or any value in between 0% and 100%. Alternatively, the change may be 1-fold, 1.5-fold, 2-fold, 3-fold, 4-fold, 5-fold or more, or any values in between 1-fold and five-fold. The change may be statistically significant with a p value of 0.1, 0.05, 0.001, or 0.0001.

(56) The term “stable disease” refers to a diagnosis for the presence of a melanoma, however the melanoma has been treated and remains in a stable condition, i.e. one that that is not progressive, as determined by imaging data and/or best clinical judgment.

(57) The term “progressive disease” refers to a diagnosis for the presence of a highly active state of a melanoma, i.e. one has not been treated and is not stable or has been treated and has not responded to therapy, or has been treated and active disease remains, as determined by imaging data and/or best clinical judgment.

EXAMPLES

(58) The disclosure is further illustrated by the following examples, which are not to be construed as limiting this disclosure in scope or spirit to the specific procedures herein described. It is to be understood that the examples are provided to illustrate certain embodiments and that no limitation to the scope of the disclosure is intended thereby. It is to be further understood that resort may be had to various other embodiments, modifications, and equivalents thereof which may suggest themselves to those skilled in the art without departing from the spirit of the present disclosure and/or scope of the appended claims.

Example 1

(59) Derivation of a 28-Marker Gene Panel

(60) Raw probe intensities (n=6,892,960 features) from n=49 whole blood samples were used to identify genes that best discriminated between different types of melanoma samples e.g., treated versus untreated, simultaneously. A total of 28 transcripts were identified in an unbiased manner as potential markers of melanoma behavior (Table 2).

(61) An artificial intelligence model of melanoma disease dynamics was built using normalized gene expression of these 28 markers in whole blood from Controls (n=90), Responders/Stable (n=68), and Progressive (n=66) samples. The dataset was randomly split into training (n=169) and testing (n=55) partitions for model creation and validation respectively. Five algorithms (XGB, RF, TreeBag, SVM, NNET) were identified that best predicted the training data. In the test set, each algorithm produced probability scores that predicted the sample. Each probability score reflects the “certainty” of an algorithm that an unknown sample belongs to Control, Responder/Stable or Progressive class. For example, an unknown sample 51 can have the following probability vector {Control=20, Responder=50, Progressive=30}. This sample would be considered a responder, given a score of 50, but there may be potential of progression (probability of 30) (FIG. 1).

(62) TABLE-US-00002 TABLE 2 Amplicon produced Chromosome using forward and Assay Exon Melanoma Biomarker or Housekeeping Gene location UniGene reverse primers Loca- Bound- Symbol Name [Cytogenetic band] ID RefSeq Length (bp) tion ary ATL1 atlastin GTPase 1 Chr.14: Hs.584905 NM_001127713.1 65 1544 12-13 50533082-50633068 [14q22.1] ATP6V0D ATPase H⁺ transporting V0 Chr.16: Hs.106876 NM_004691.4 72 224 1-2 subunit d1 67438014-67481186 [16q22.1] C1ORF21 chromosome 1 open Chr.1: Hs.497159 NM_030806.3 84 805 5-6 reading frame 21 184387016-184629021 [1q25.3] CFLAR CASP8 and FADD like Chr.2: Hs.390736 NM_001127183.2 59 874 3-4 apoptosis regulator 201116015-201172688 [2q33.1] CFLAR-AS1 CFLAR antisense RNA 1 Chr.2: Hs.664613 NR_040030.1 103 883 6-7 201140289-201157792 [2q33.1] CHP1 calcineurin like EF-hand Chr.15: Hs.406234 NM_007236.4 120 212 1-2 protein 1 41231149-41281887 [15q15.1] DDX55 DEAD-box helicase 55 Chr.12: Hs.286173 NM_020936.2 74 183 1-2 123602077-123620943 [12q24.31] DMD dystrophin Chr.X: Hs.495912 NM_000109.3 76 9607 63-64 31119219-33339609 [Xp21.1] DNAJC9 DnaJ heat shock protein Chr.10: Hs.408577 NM_015190.4 102 198 1-1 family (Hsp40) member C9 73241954-73247331 [10q22.2] ENOSF1 enolase superfamily Chr.18: Hs.658550 NM_001126123.3 110 X 13-14 member 1 670012-712664 [18p11.32] FANCL Fanconi anemia Chr.2: Hs.631890 NM_001114636.1 138 540 6-7 complementation group L 58159243-58241380 [2p16.1] HJURP Holliday junction Chr.2: Hs.532968 NM_018410.4 52 399 4-5 recognition protein 233836701-233854566 [2q37.1] HLA-DOA major histocompatibility Chr.6: Hs.631991 NM_002119.3 124 160 1-2 complex, class II, DO alpha 33004182-33009612 [6p21.32] HLA-DRA major histocompatibility Chr.6: Hs.520048 NM_019111.4 129 884 4-5 complex, class II, DR alpha 32439842-32445046 [6p21.32] HNRNPA3P1 heterogeneous nuclear Chr.10: Hs.632956 NR_002726.2 99 2455 1-1 ribonucleoprotein A3 43787412-43790417 pseudogene 1 [10q11.21] IL23A interleukin 23 subunit alpha Chr.12: Hs.382212 NM_016584.2 107 323 1-2 56334159-56340410 [12q13.3] IQGAP1 IQ motif containing Chr.15: Hs.430551 NM_003870.3 69 4562 34-35 GTPase 90388241-90502243 activating protein 1 [15q26.1] LOC494127 NFYC pseudogene Chr.9: Hs.626316 NR_036691.1 86 150 1-1 87083886-87085225 [9q21.33] LOC646471 uncharacterized Chr.1: Hs.727271 NR_024498.1 80 2858 1-1 LOC646471 25819954-25823606 [1p36.11] LOH12CR loss of heterozygosity, 12, Chr.12: Hs.67553 NR_024061.1 69 140 1-2 chromosomal region 2 12355408-12357067 (non-protein coding) [12p13.2] MTRNR2L2_MTO1 CUSTOM PRIMER PBXIP1 PBX homeobox interacting Chr.1: Hs.505806 NM_020524.3 60 189 2-3 protein 1 154944080-154956163 [1q21.3] RNF5 ring finger protein 5 Chr.6: Hs.731774 NM_006913.3 67 273 1-2 32178385-32180793 [6p21.32] SERTAD2 SERTA domain containing Chr.2: Hs.591569 NM_014755.2 91 288 1-2 2 64631621-64751091 [2p14] SLC35G5 solute carrier family 35 Chr.8: Hs.458397 NM_054028.1 144 541 1-1 member G5 11330986-11332186 [8p23.1] SPATS2L spermatogenesis associated Chr.2: Hs.120323 NM_001100422.1 89 1364 10-11 serine

rich 2 like 200305881-200482263 [2q33.1] TDRD7 tudor domain containing Chr.9: Hs.193842 NM_001302884.1 59 3084 15-16 7 97412020-97496125 [9q22.33] TOX4 TOX high mobility group Chr.14: Hs.555910 NM_001303523.1 145 447 3-3 (housekeeping box family member 4 21477176-21499177 gene) [14q11.2] TPT1 tumor protein, Chr.13: Hs.374596 NM_001286272.1 131 377 3-3 (housekeeping translationally-controlled 1 45333471-45341284 gene) [13q14.13] TXK TXK tyrosine kinase Chr.4: Hs.479669 NM_003328.2 113 1265 11-12 48066393-48135322 [4p12] YY2 YY2 transcription factor Chr.X: Hs.673601 NM_206923.3 110 552 1-1 21855987-21858727 [Xp22.12]

Diagnosis: Identification of Samples as Melanoma

(63) In the test set 1, the data for the utility of the test to differentiate melanoma from controls are included in Table 3. The metrics are included in FIG. 2. These are: sensitivity >90%, specificity 100%, PPV 100%, NPV 87%. The overall accuracy is 94%. The tool can therefore differentiate between controls and aggressive and stable melanoma disease.

(64) TABLE-US-00003 TABLE 3 Confusion matrix showing classification accuracy of the 5-model algorithm that determines whether a sample is a melanoma or a control in blood samples

Predicted/Reference	Melanoma	Control
Melanoma	30	0
Control	20	3

Sensitivity 91% Specificity 100% Positive Predictive Value 100% Negative Predictive Value 87% Accuracy 94%

(65) The test was evaluated in a second test set (test set 2) that included 74 controls and 92 melanoma patients. The mean Melanomx score in the melanoma group was 73 ± 31 versus 10 ± 8 in the control group (FIG. 3A). The receiver operator curve analysis demonstrated the score exhibited an area under the curve (AUC) of 0.96 (FIG. 3B) and the metrics were 88-100% (FIG. 3C).

(66) Correlation with Surgical Removal of Melanoma.

(67) In the surgical series (n=46), removal of melanoma decreased the score from 74 ± 16 to 20 ± 12 (FIG. 4A). Evaluation of the post-surgical group identified the score was significantly different between patients with no evidence of disease 13 ± 6 (not different to the control group) and those with residual disease (33 ± 9) (FIG. 4B). The Melanomx score can therefore define the extent of surgery and identify those who are surgical cures.

(68) Evaluation of Immune-Therapy and Targeted (BRAF Inhibitor) Therapy.

(69) In the therapy series (n=30), treatment with ipilimumab (immune therapy targeting CTLA-4), nivolumab (immune therapy targeting PD-1) or a BRAF inhibitor (vemurafenib) for 5 months significantly decreased the Melanomx scores from 88 ± 12 to 15 ± 5 ($p < 0.0001$), 35 ± 14 ($p < 0.001$) and 38 ± 21 ($p < 0.0001$), respectively (FIG. 5). All patients in the treated groups were stable. This indicates that the Melanomx score can therefore be used to measure the efficacy of both immune and targeted-therapy in melanoma and that a decrease in score correlated with response to therapeutic intervention.

(70) Confirmation that Gene Expression is Melanoma Derived.

(71) We confirmed that melanoma was the source for the blood-based gene expression assay by evaluating expression in different melanoma cell lines, in tumor tissue and by comparing expression in blood with tumor tissue collected at the same time-point during surgery.

(72) All 28 genes were highly expressed in all 3 melanoma cell lines. Scores ranged from 94 ± 6 (Hs294T) to 82 ± 5 (CHL) to 42 ± 9 (A375) (FIG. 6A).

(73) Spike-in experiments using these 3 cell lines and normal whole blood demonstrated that gene expression scores were detected when as few as 1 cell was spiked into 1 ml of blood. One single melanoma cell was detectable. Scores ranged from 21 ± 4 (A375) to 30 ± 5 (CHL) to 34 ± 2 (H2294) (FIG. 6B). Scores were significantly elevated compared to control blood (no spike-in; $p < 0.0001$).

(74) We then evaluated gene expression in tumor tissue. All 28 genes were highly expressed in melanoma compared to matched normal tissue and scores were significantly elevated 70 ± 20 versus 4 ± 3 ($p < 0.0001$) (FIG. 7A).

(75) We also compared gene expression in tumor tissue and blood collected at the same time point. Gene expression was highly concordant (Pearson r: 0.74-0.83, median: 0.819) identifying gene

expression in tumor tissue and blood was concordant (FIG. 7B).

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EQUIVALENTS

(77) While the present invention has been described in conjunction with the specific embodiments set forth above, many alternatives, modifications and other variations thereof will be apparent to those of ordinary skill in the art. All such alternatives, modifications and variations are intended to fall within the spirit and scope of the present invention.

Claims

1. A method of treating a melanoma in a human subject in need thereof, the method comprising: a) determining the expression level of 29 biomarkers from a biological fluid sample from the subject by contacting the test sample with a plurality of agents specific to detect the expression of the 29 biomarkers, wherein the 29 biomarkers comprise each of atlastin GTPase 1 (ATL1), ATPase H+ transporting V0 subunit d1 (ATP6V0D), chromosome 1 open reading frame 21 (C1ORF21), CASP8 and FADD like apoptosis regulator (CFLAR), CFLAR antisense RNA 1 (CFLAR-AS1), calcineurin like EF-hand protein 1 (CHP1), DEAD-box helicase 55 (DDX55), dystrophin (DMD), DnaJ heat shock protein family (Hsp40) member C9 (DNAJC9), enolase superfamily member 1 (ENOSF1), Fanconi anemia complementation group L (FANCL), Holliday junction recognition protein (HJURP), major histocompatibility complex, class II, DO alpha (HLA-DOA), major histocompatibility complex, class II, DR alpha (HLA-DRA), heterogeneous nuclear ribonucleoprotein A3 pseudogene 1 (HNRNPA3P1), interleukin 23 subunit alpha (IL23A), IQ motif containing GTPase activating protein 1 (IQGAP1), NFYC pseudogene (LOC494127), uncharacterized LOC646471 (LOC646471), loss of heterozygosity, 12, chromosomal region 2 (non-protein coding) (LOH12CR), PBX homeobox interacting protein 1 (PBXIP1), ring finger protein 5 (RNF5), SERTA domain containing 2 (SERTAD2), solute carrier family 35 member G5 (SLC35G5), spermatogenesis associated serine rich 2 like (SPATS2L), tudor domain containing 7 (TDRD7), TXK tyrosine kinase (TXK), YY2 transcription factor (YY2), and at least one housekeeping gene; b) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of the at least one housekeeping gene, thereby obtaining a normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; c) inputting each normalized expression level into an algorithm to generate a score, wherein the algorithm is the product of a trained classifier built from at least one predictive classification algorithm; d) determining that the human subject has melanoma by determining that the score is greater than or equal to a first predetermined cutoff value, wherein the first predetermined cutoff value is 20 on a scale of 0 to 100; and e) administering to the human subject at least one

- therapeutically effective amount of at least one of: ##STR00002## or any combination thereof.
2. The method of claim 1, wherein the method comprises administering at least one therapeutically effective amount of: ##STR00003##
3. The method of claim 1, wherein the method comprises administering at least one therapeutically effective amount of: ##STR00004##
4. The method of claim 1, wherein the method comprises administering at least one therapeutically effective amount of: ##STR00005##
5. The method of claim 1, wherein the method comprises administering at least one therapeutically effective amount of: ##STR00006##
6. The method of claim 1, wherein the at least one housekeeping gene comprises TOX high mobility group box family member 4 (TOX4), tumor protein, translationally-controlled 1 (TPT1), or any combination thereof.
7. The method of claim 6, wherein the at least one housekeeping gene comprises TOX4 and TPT1.
8. The method of claim 7, wherein step (b) comprises: i) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of TOX4, thereby obtaining a first normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; ii) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of TPT1, thereby obtaining a second normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; and iii) averaging the first normalized expression level and the second normalized expression level, thereby obtaining a normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2.
9. The method of claim 1, wherein at least one of the 32 biomarkers is RNA, cDNA, or protein.
10. The method of claim 9, wherein when the biomarker is RNA, the RNA is reverse transcribed to produce cDNA, and the produced cDNA expression level is detected.
11. The method of claim 1, wherein the expression level of the biomarker is detected by forming a complex between the biomarker and a labeled probe or primer.
12. The method of claim 9, wherein when the biomarker is protein, the protein is detected by forming a complex between the protein and a labeled antibody.
13. The method of claim 10, wherein when the biomarker is RNA or cDNA, the RNA or cDNA is detected by forming a complex between the RNA or cDNA and a labeled nucleic acid probe or primer.
14. The method of claim 13, wherein the label is a fluorescent label.
15. The method of claim 13, wherein the complex between the RNA or cDNA and the labeled nucleic acid probe or primer is a hybridization complex.
16. The method of claim 1, wherein the algorithm is XGB, RF, glmnet, cforest, CART, treebag, knn, nnet, SVM-radial, SVM-linear, NB, NNET, mlp, or logistic regression modeling.
17. The method of claim 1, wherein the algorithm comprises a model of melanoma disease

dynamics built using the expression levels of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 from reference blood samples from a plurality of subjects having melanoma and a plurality of subjects not having melanoma.
